A System for Acquisition of Noun Concepts from Utterances for Images Using the Label Acquisition Rules

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Abstract. This paper presents a system which acquires noun concepts (labels for images) based on infant vocabulary acquisition models. In order to improve its performance, we introduced "label acquisition rules" into this system.

1 Introduction

Various types of robots to entertain people have been developed in recent years [1][2][3][4]. These robots often have a dialogue capability, but people are not satisfied at their communication skills. The robots should be capable of learning new knowledge and have a higher capability to keep people interested over time.

We developed a **S**ystem for **N**oun **C**oncepts **A**cquisition from utterances for **I**mage, called SINCA in short (which means "evolution" in Japanese). It is a language acquisition system without knowledge of grammar and vocabulary, which learns noun concepts from user inputs. The system was not sufficient in terms of usage of knowledge from experience, therefore we introduced a "label acquisition rule" into it. By doing so, SINCA can learn phrases that tend to contain noun concepts.

Currently, SINCA is for Japanese only. The language acquisition method of this system is very simple. The method is independent of language rules. Therefore SINCA is expected to work successfully while using any language.

2 Algorithm of SINCA

2.1 Input

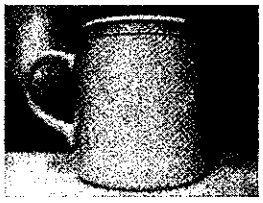
A user input consists of an image paired with a spoken utterance. Figure 1 shows some examples of inputs.[1]

First, a user chooses an object O which he or she likes and captures its image with a web camera. A user has to try to capture the whole object O in the image.

[1] Strings indicated in boldface are labels and in italic are Japanese.

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Next, a user imagines an utterance that an infant might be exposed to when listening to caregivers while gazing at the object O in the environment. A user enters the utterance on the keyboard as a linguistic input. SINCA does not carry out morphological analysis for the linguistic input, because we think infant capacity for word segmentation is not perfect[5].



*Kore-ha **KAPPU**-tte iu-n-da-yo.*
(This is a cup.)
***KAPPU**-ni gyūnyū ireyōka.*
(Let's pour some milk into the cup.)

Fig. 1. Examples of input data

2.2 Image Processing

We use the "ERSP vision" included in the ERSP 3.1 Software Development Kit[6] for image processing. The ERSP Vision enables a robot or device to recognize 2D and 3D objects in real world settings where lighting and placement are not controlled. It tells whether the object in the present input image appears in the previous input images or not.

2.3 Common Parts and Basic Score

When a user inputs an image of the object O and an utterance, the system extracts all sections of a string that matches with the previous input utterances with the image of the same object O. We call the strings common parts. After this process, the system deals with them as candidates of a label for the object O.

The system provides every common part with a "basic score". The basic score is based on frequency of appearance and number of characters, and means how proper for the label the common part is. The basic score is defined as follows:

$$SCORE = \alpha \times \frac{F}{PN} \times \sqrt{L} \tag{1}$$

where, α is a coefficient which reduces the basic score if the common part has appeared with other object than O, F is frequency of appearance of the common part with the images of O, PN is the number of times that the user inputs with the images of O, and L is the number of characters of the common part.

2.4 Output

If the system finds a common part which basic score exceeds a threshold, outputs it as text. The reason for doing this comes from the belief that there is a high possibility that such common parts are proper for a label.

A user evaluates an output by choosing one of the following keywords:

- Good : It is proper for the label.
- Almost : It makes some sense but is not proper for the label.
- Bad : It makes no sense.

The system recalculates the basic score based on the keyword chosen by the user.

2.5 Acquisition of the Noun Concepts

After repeating these processes, if there is a common part which score is more than 30.0 and which has been rated as "Good", the system acquires the common part as the label for *O*.

2.6 Label Acquisition Rules

Humans can put their newfound knowledge to learn their native language effectively. This system imitates humans' way with "label acquisition rules".

A label acquisition rule is like a template, which enables recursive learning for acquisition of noun concepts. The system generates the label acquisition rules after acquisition of a label. When the system acquires a string S as a label for an object, the system picks up the previous linguistic inputs with the images of the object which contain the string S. Then, the system replaces the string S in the linguistic inputs with a variable "@". These abstracted sentences are called the label acquisition rules. To give an actual example, if an acquired label is "***WAN-CHAN*** (a doggy)" and a previous input is "*Atchi-ni* ***WAN-CHAN****-ga iru-yo.* (There is a doggy over there.)", then a label acquisition rule is "*Atchi-ni @-ga iru-yo.* (There is @ over there.)".

If there are matching rules with linguistic inputs, the system extracts strings which are proper for variable "@". The scores of extracted strings are increased.

3 Evaluation Experiment of SINCA with Label Acquisition Rules

3.1 Experimental Procedure

We compare a system using label acquisition rules with the one not using them. A user inputs the same data into both systems until they acquire 30 labels. The linguistic input data were collected by a questionnaire from 31 people. 1001 sentences were collected. We use 396 containing words representing labels for objects.

3.2 Experimental Results and Considerations

The system acquires 85 types of rules until it acquires 30 labels. Of them, only 13 rules are applied of 74 times in total. The system with the rules needs 5.6

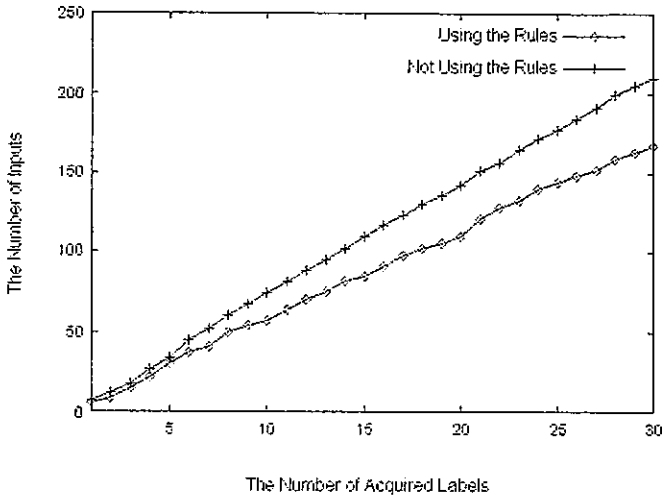


Fig. 2. The relation between number of inputs and acquisition of labels

inputs on average to acquire one label. In contrast, the system without the rules needs 7.0 inputs on an average. Figure 2 shows how many inputs the systems need to acquire labels.

This result indicates that using them makes the acquisition of labels more efficient. A lot of rules are acquired, but only 15.3% of those are applied. This means that a small proportion of the highly capable rules are applied. We should therefore consider deleting unnecessary rules.

4 Experiment to Evaluate the Generality of Label Acquisition Rules

Infants do not hear only infant-directed speech but also a TV set playing, conversations between adults, and so on. On the contrary, the user inputs for SINCA are always utterances that adults direct to it. SINCA uses web-based information to make up for this difference.

We get generalities of the label acquisition rules based on the results of Google search[7]. By these generalities we confirm if deleting unnecessary rules is possible.

4.1 Experimental Procedure

We search for every rule on Google. Variables in the rules are replaced by wildcards. To give an actual example, if an acquired rule is "*Kore-ha @ da-yo* (This is a @.)", a search query is "*Kore-ha * da-yo* (This is a *)".

4.2 Experimental Results and Considerations

Figure 3 shows relationship between the results of Google search and the number of rules. There are 41 rules where Google search returns 0 hits. It is nearly half